

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CELL, The Unit of Life

LECTURE-10

ZOOLOGY

DAVOOD SIR (CEO of NG)



Topics To Be Covered

1 **EXERCISE**

② NEET PYQ

Question: 1

Ex 20



Which of the following is not correct?

- 1 Robert Brown discovered the cell. *F*
- 2 Schleiden and Schwann formulated the cell theory. *T*
- 3 Virchow explained that cells are formed from pre-existing cells. *T*
- 4 A unicellular organism carries out its life activities within a single cell *T*

Question: 2

Exe.



New cells generate from

1 bacterial fermentation X

2 regeneration of old cells X

3 pre-existing cells ✓

4 abiotic materials X

Question: 3

Ex 28



Match the following
Column I

(a) Cristae

(b) Cisternae

(c) Thylakoids

Column II

(i) Flat membranous sacs in stroma

(ii) Infoldings in mitochondria

(iii) Disc-shaped sacs in Golgi apparatus

a(ii) b(iii) c(i)

Question: 4

EXER



Which of the following is correct:

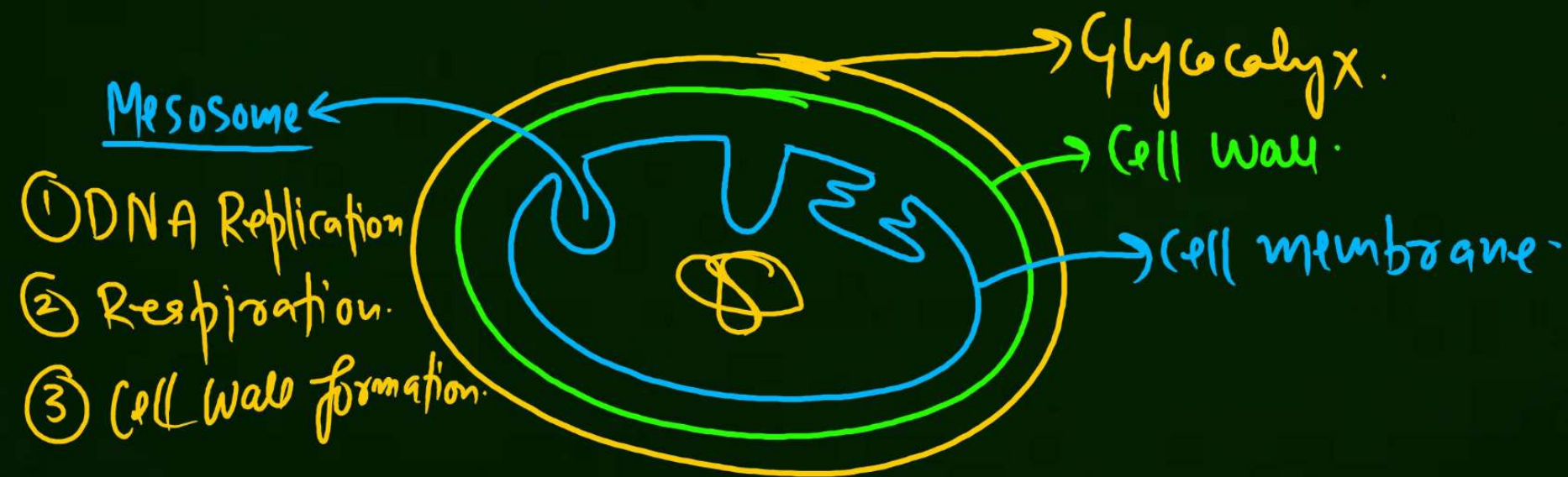
- 1 Cells of all living organisms have a nucleus. ✗
- 2 Both animal and plant cells have a well defined cell wall. ✗
- 3 In prokaryotes, there are no membrane bound organelles. ✓
- 4 Cells are formed de novo from abiotic materials. ✗

Question: 5

Exer.



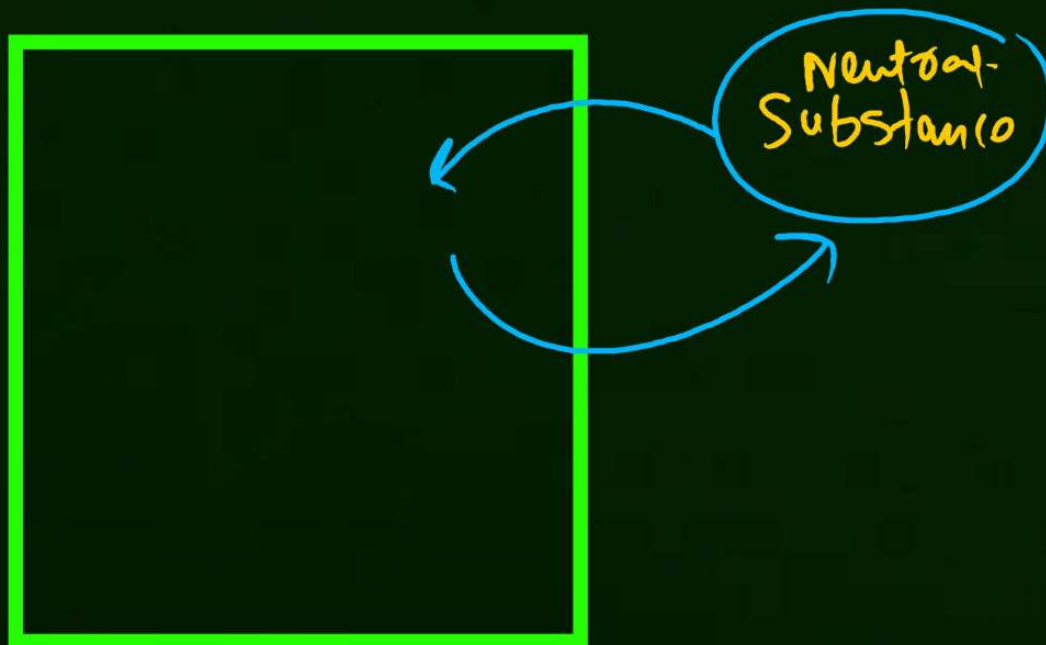
What is a mesosome in a prokaryotic cell? Mention the functions that it performs.



Question: 6

H/w.

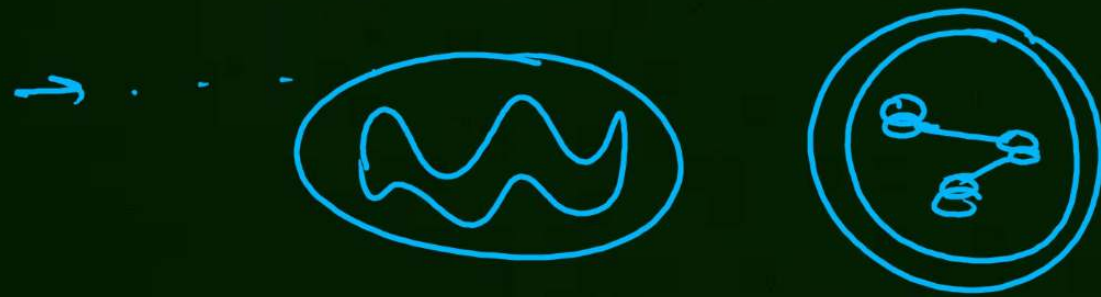
How do neutral solutes move across the plasma membrane? Can the polar molecules also move across it in the same way? If not, then how are these transported across the membrane?



Question: 7

Name two cell-organelles that are double membrane bound. What are the characteristics of these two organelles? State their functions and draw labelled diagrams of both.

Ans: → Mitochondria & Plastid. (Semiautonomous cell organelles).



Question: 8

What are the characteristics of prokaryotic cells? *Bacterial cells.*

Question: 9

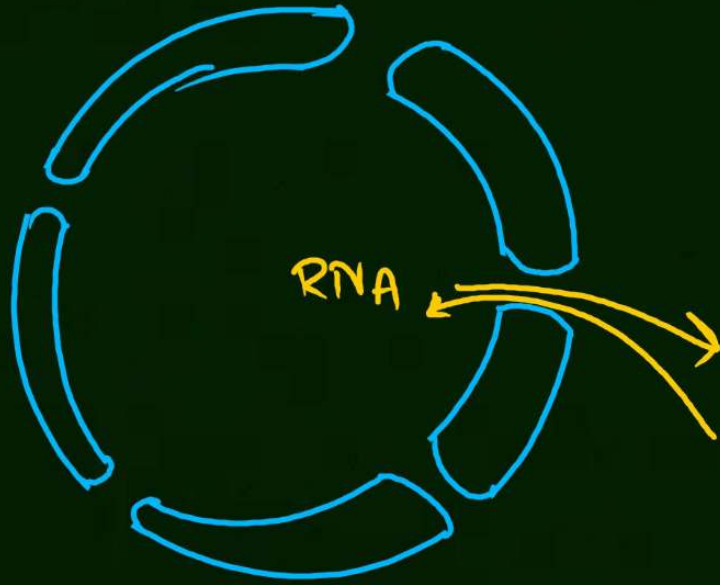
Multicellular organisms have division of labour. Explain.

Question: 10

Cell is the basic unit of life. Discuss in brief.

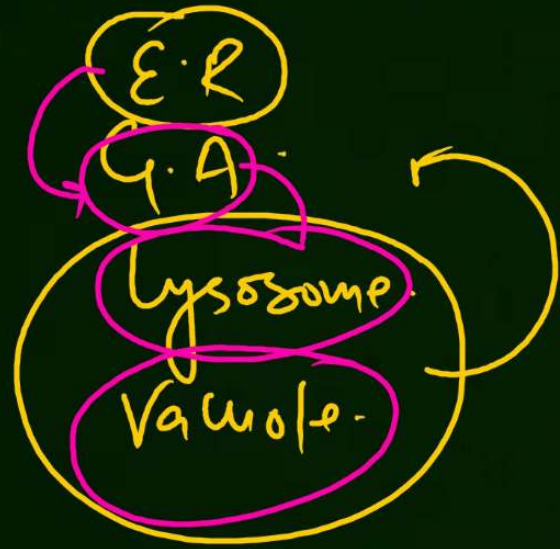
Question: 11

What are nuclear pores? State their function.



Question: 12

Both lysosomes and vacuoles are endomembrane structures, yet they differ in terms of their functions. Comment.



Question: 13

Describe the structure of the following with the help of labelled diagrams.

(i) Nucleus

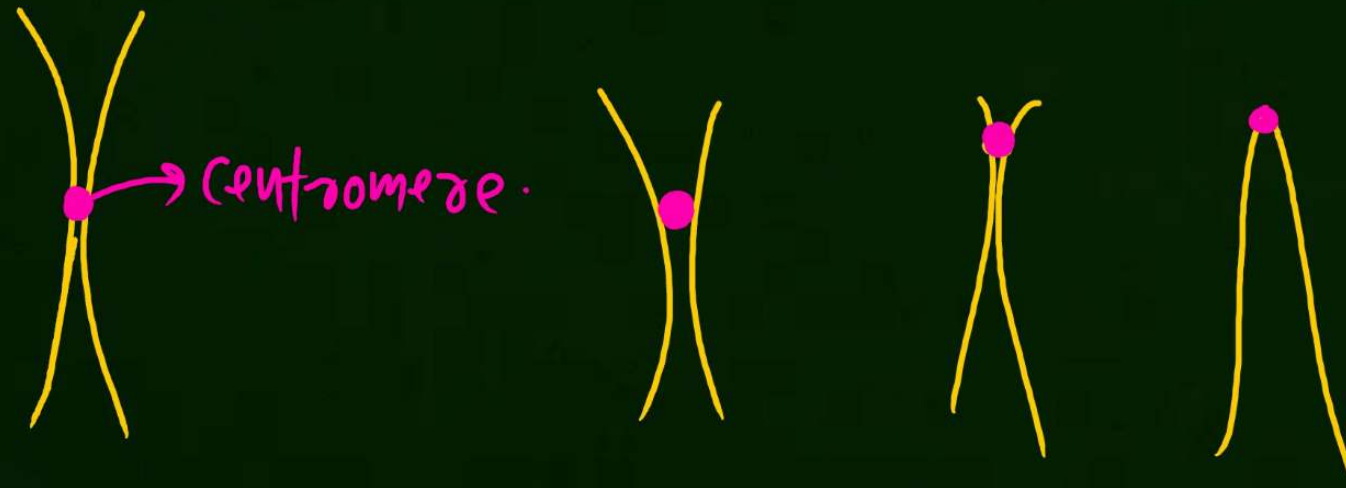
Diagram

(ii) Centrosome

Diagram

Question: 14

What is a centromere? How does the position of centromere form the basis of classification of chromosomes. Support your answer with a diagram showing the position of centromere on different types of chromosomes.



Question: 15

The concept of "Omnis cellula-e-cellula" regarding cell division was first proposed by [NEET-2019]

- 1 Aristotle
- 2 Rudolf Virchow.
- 3 Theodore Schwann
- 4 Schleiden

Question: 16

Names of Schleiden and Schwann are associated with: [NEET-1993]

- 1 Protoplasm as the physical basis of life
- 2 Cell theory.
- 3 Theory of cell lineage
- 4 Nucleus functions as control centre of cell

Question: 17

The plasma membrane consists mainly of

[NEET-2010]

- 1 Phospholipids embedded in a protein bilayer
- 2 Proteins embedded in a phospholipid bilayer. ✓
- 3 Proteins embedded in a polymer of glucose molecules
- 4 Proteins embedded in a carbohydrate bilayer.

Question: 18

Select the correct statement from the following regarding cell membrane.
[NEET-1993]

- 1 Na^+ and K^+ ions move across cell membrane by passive transport 
- 2 Proteins make up 60 to 70% of the cell membrane. 
- 3 Lipids are arranged in a bilayer with polar heads towards the inner part. 
- 4 Nucleus functions as control centre of cell 

Question: 19

HW



Keeping in view the 'fluid mosaic model' for the structure of cell membrane, which one of the following statement is correct with respect to the movements of lipids and proteins from one lipid monolayer to the other (described as flip-flop movement)?

[NEET-1993]

- 1 While proteins can flip-flop, lipids can not
- 2 Neither lipids, nor proteins can flip-flop
- 3 Both lipids and proteins can flip-flop
- 4 While lipids can rarely flip-flop, proteins cannot.

Question: 20

Which one of the following is not a constituent of cell membrane?
[NEET-2007]

1 Glycolipids ✓

2 Proline. ✗

3 Phospholipids ✓

4 Cholesterol ✓

Question: 21

H/W.



According to widely accepted "fluid mosaic model" cell membranes are semi-fluid, where lipids and integral proteins can diffuse randomly. In recent years, this model has been modified in several respects. In this regard, which of the following statements is incorrect? [NEET-1993]

- 1 Proteins in cell membranes can travel within the lipid bilayer.
- 2 Proteins can also undergo flip-flop movements in the lipid bilayer
- 3 Proteins can remain confined within certain domains of the membrane
- 4 Many proteins remain completely embedded within the lipid bilayer

Question: 22

The organelles that are included in the endomembrane system are
[NEET-1993]

1 Golgi complex, endoplasmic reticulum, mitochondria and lysosomes ✗

2 Endoplasmic reticulum, mitochondria, ribosomes and lysosomes ✗

3 Endoplasmic reticulum, Golgi complex, lysosomes and vacuoles. ✓

4 Golgi complex, mitochondria, ribosomes and lysosomes ✗

Question: 23

Which of the following statements with respect to endoplasmic reticulum is incorrect? [NEET-2002]

- 1 RER has ribosomes attached to ER ✓
- 2 SER is devoid of ribosomes ✓
- ~~3~~ In prokaryotes only RER are present. ✗
- 4 SER are the sites for lipid synthesis ✓

Question: 24

Which of the following statements are correct with respect of Golgi apparatus?

- A. It is the important site of formation of glycoprotein and glycolipids. ✓
- B. It produces cellular energy in the form of ATP. ✗
- C. It modifies the protein synthesised by ribosomes on ER. ✓
- D. It facilitates the transport of ions. ✗
- E. It provides mechanical support. ✗

Choose the most appropriate answer from the options given below. [NEET-1993]

1 B and C only

2 A and C only. ✓

3 A and D only

4 D and E only

Question: 25

Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells?
[NEET-2020]

1 Endoplasmic reticulum ✗

2 Peroxisomes ✗

3 Golgi bodies ✓

4 Polysomes ✗

Question: 26

$E.R \rightarrow G.A \rightarrow Ly \rightarrow$

Which of the following statements is not correct?

[NEET-2019]

- 1 Lysosomes are formed by the process of packaging in the endoplasmic reticulum. ✗
- 2 Lysosomes have numerous hydrolytic enzymes ✓
- 3 The hydrolytic enzymes of lysosomes are active under acidic pH ✓
- 4 Lysosomes are membrane-bound structures ✓

Question: 27

Assertion (A): The primary function of the Golgi apparatus is to package materials made by the endoplasmic reticulum and deliver them to intracellular targets and outside the cell. *T.*

Reason (R): Vesicles containing materials made by the endoplasmic reticulum fuse with the cis face of the Golgi apparatus, and they are modified and released from the trans face of the Golgi apparatus. *T.*

[NEET-2025]

- 1 A is false but R is true
- ☒ 2 Both A and R are true AND R correctly explains A.
- 3 Both are true but R does not explain A
- 4 A is true but R is false

Question: 28

Cell wall shows-

[NEET-1991]

1 Complete permeability. ✓

2 Semi permeability ✗

3 Differential permeability ✗

4 Impermeability ✗

Question: 29

Which one of the following structures between two adjacent cells is an effective transport pathway?
[NEET-2010]

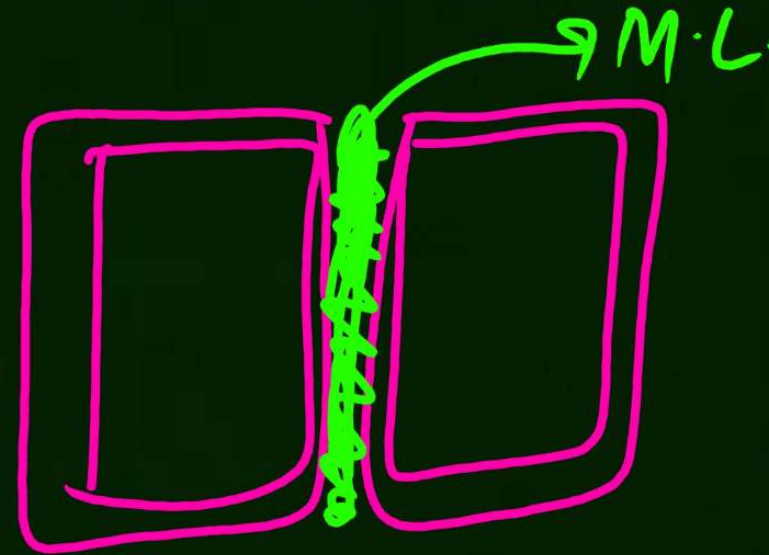
- 1 Plasmodesmata 
- 2 Plastoquinones
- 3 Endoplasmic reticulum
- 4 Plasmalemma

Question: 30

Middle lamella is composed mainly of

[NEET-2009]

- 1 Muramic acid
- 2 Calcium pectate ✓
- 3 Phosphoglycerides
- 4 Hemicellulose



Question: 31

Plasmodesmata are-

[NEET-2009]

- 1 Locomotory structures ✗
- 2 Membranes connecting the nucleus with plasmalemma ✗
- 3 Connections between adjacent cells ✓
- 4 Lignified cemented layers between cells. ✗

Question: 32

Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP?
[NEET 2017]

1 Ribosome

2 Chloroplast

3 Mitochondrion ✓

4 Lysosome

Question: 33

Which of the following statements regarding mitochondria is incorrect? [NEET-2019]

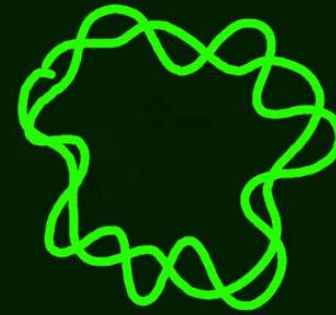
- 1 Mitochondrial matrix contains single circular DNA molecule and ribosomes. ✓
- 2 Outer membrane is permeable to monomers of carbohydrates, fats and proteins. ✓
- 3 Enzymes of electron transport are embedded in outer membrane. ✗
- 4 Inner membrane is convoluted with infoldings. ✓

Question: 34

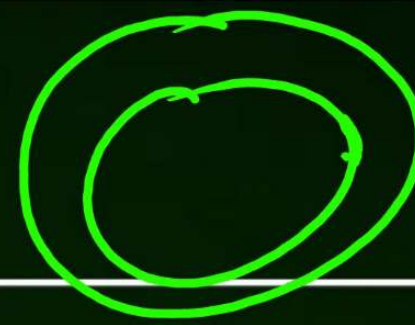
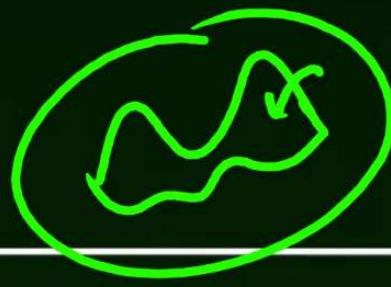
The DNA present in chloroplast is

[NEET-2024]

- 1** linear, double stranded
- 2** circular, double stranded ✓
- 3** linear, single stranded
- 4** circular, single stranded.



Question: 35



Given below are two statements:

Statements I: Mitochondria and chloroplast both double membrane bound organelles. *T.*

Statements II: Inner membrane of mitochondria is relative less permeable, as compared chloroplast. *F*

In the light of the above statements, choose the appropriate answer from the options given below. [NEET 2024]

- 1** Both Statement I and Statement II are correct.
- 3** Both Statement I and Statement II incorrect.
- 2** Statement I is correct but Statement II incorrect.
- 4** Statement I is incorrect but Statement II incorrect.

Question: 36

Match List – I with List – II.

List – I

(p) Cristae

(q) Thylakoids

(r) Centromere

(s) Cisternae

List – II

(i) Primary constriction in chromosome

(ii) Disc-shaped sacs in Golgi apparatus

(iii) Infoldings in mitochondria

(iv) Flattened membranous sacs in stroma of plastids

Choose the correct answer from the options given below

[NEET-2021]

1 (p)-(ii), (q)-(iii), (r)-(iv), (s)-(i)

2 (p)-(iv), (q)-(iii), (r)-(ii), (s)-(i)

3 (p)-(i), (q)-(iv), (r)-(iii), (s)-(ii)

4 (p)-(iii), (q)-(iv), (r)-(i), (s)-(ii)

Question: 37

Match List-I with List-II.

Column-I

- A. Golgi apparatus
- B. Lysosomes
- C. Vacuoles
- D. Ribosomes

Column-II

- I. Synthesis of protein
- II. Trap waste and excretory products
- III. Formation of glycoproteins and glycolipids
- IV. Digesting biomolecules

Choose the correct answer from the options given below

[NEET 2019]

~~1~~ A-(iii), B-(iv), C-(ii), D-(i)

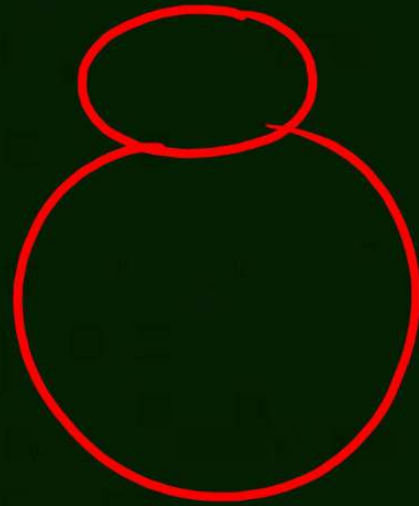
2 A-(iv), B-(iii), C-(i), D-(ii)

3 A-(iii), B-(ii), C-(iv), D-(i)

4 A-(i), B-(ii), C-(iv), D-(iii)

Question: 38Class 12th.

How many different proteins does the [?]ribosome consist of?
[NEET-2023]
Eukaryotic

1 40**2** 20**3** 80**4** 100RNA.
Protein.

Question: 39

80's $\rightarrow$ 60's, 40's
70's $\rightarrow$ 50's, 30's



From the statements given below choose the correct option:

- A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S. T.
- B. Each ribosome has two sub-units. T.
- C. The two sub-units of 80S ribosome are 60S and 40S while that of 70S are 50S and 30S. T.
- D. The two sub-units of 80S ribosome are 60S and 20S and that of 70S are 50S and 20S. F.
- E. The two sub-units of 80S are 60S and 30S and that of 70S are 50S and 30S. F.

[NEET-2025]

☒ 1 A, B, C are true

☐ 2 A, B, D are true

☐ 3 A, B, E are true

☐ 4 B, D, E are true

Question: 40

How many different proteins does the ribosome consist of?

[NEET-2023]

1 40

2 20

3 80 ✓

4 100

Question: 41

Match List-I with List-II.

Column-I

- A. Nucleolus
- B. Centriole
- C. Leucoplasts
- D. Golgi apparatus

Column-II

- I. Site of formation of glycolipid
- II. Organization like the cartwheel
- III. Site for active ribosomal RNA synthesis
- IV. For storing nutrients

Choose the correct answer from the options given below

[NEET-2019]

1 A-(III), B-(II), C-(IV), D-(I)

2 A-(II), B-(III), C-(I), D-(IV)

3 A-(III), B-(IV), C-(II), D-(I)

4 A-(I), B-(II), C-(III), D-(IV)

Question: 42

Which of the following functions is carried out by cytoskeleton in a cell?
[NEET-2023]

1

Motility



2

Transportation



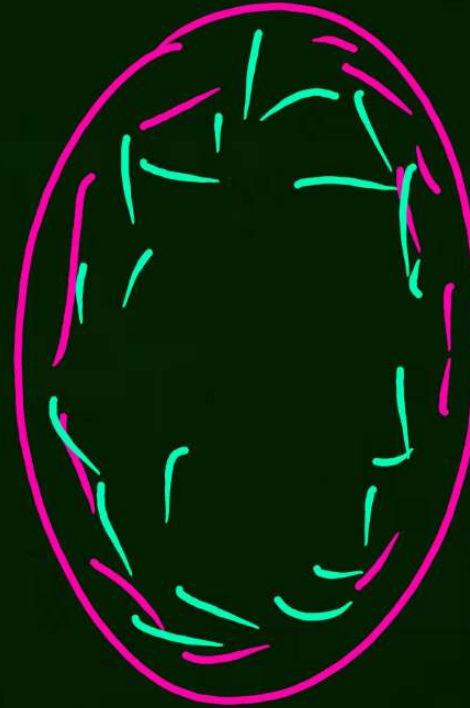
3

Nuclear division



4

Protein synthesis



Question: 43

Match List-I with List-II.

Column-I

A. Axoneme

B. Cartwheel pattern

C. Crista

D. Satellite

Column-II

I. Centriole

II. Cilia and flagella

III. Chromosome

IV. Mitochondria

Choose the correct answer from the options given below

[NEET-2024]

1

A-(IV), B-(III), C-(II), D-(I)

2

A-(IV), B-(II), C-(III), D-(I)

3

A-(II), B-(IV), C-(I), D-(III)

4

A-(II), B-(I), C-(IV), D-(III)

Question: 44

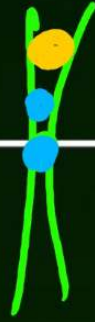
Match List-I with List-II.
Column-I

A. Metacentric chromosome

B. Acrocentric chromosome

C. Sub-metacentric

D. Telocentric chromosome



I.

II.

III.

IV.

Column-II

Centromere situated close to the end forming one extremely short and one very long arm

Centromere at the terminal end

Centromere in the middle forming two equal arms of chromosomes

Centromere slightly away from the middle forming one shorter arm and one longer arm.

Choose the correct answer from the options given below

[NEET 2022]

1 A-(III), B-(I), C-(IV), D-(II)

2 A-(I), B-(III), C-(II), D-(IV)

3 A-(II), B-(III), C-(IV), D-(I)

4 A-(I), B-(II), C-(III), D-(IV)

Question: 45

Which of the following is an incorrect statement?

[NEET-2021]

1 Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm. *T*

2 Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles. *F*

3 Microbodies are present both in plant and animal cells. *T*

4 The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm. *T*



Question: 46

When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as [NEET-2021]

1 Acrocentric

2 Metacentric

3 Telocentric

4 Sub-metacentric.



Question: 47

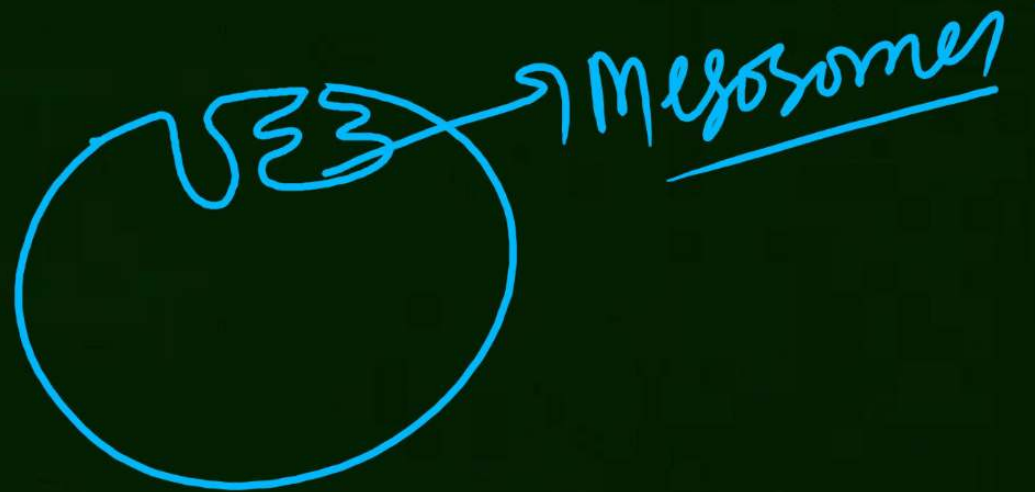
Which specialised structure in a prokaryotic cell helps in cell wall formation, DNA replication and respiration? [NEET-2025]

1 Endoplasmic reticulum

2 Mesosome ✓

3 Chromatophores

4 Cristae



Question: 48

Given below are two statements.

Statement I: In bacteria, the mesosomes are formed by the extensions of plasma membrane. ^T

Statement II: The mesosomes, in bacteria, help in DNA replication and cell wall formation. ^T

In the light of the above statements, choose the most appropriate answer from the options given below.

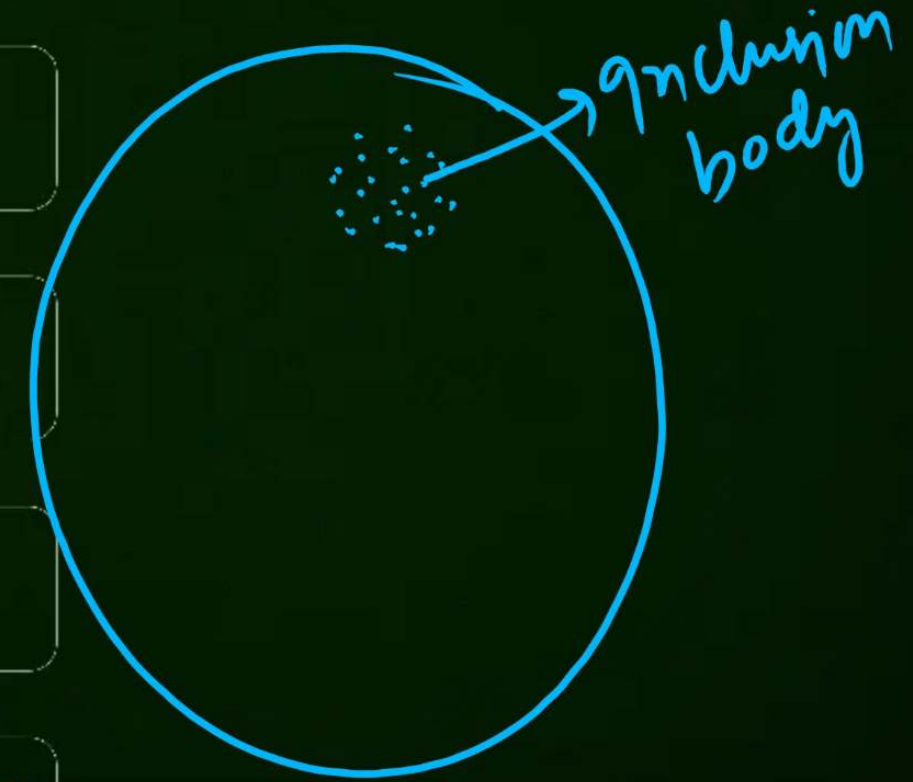
[NEET-2023]

- 1 Both Statement I and Statement II are correct. 
- 3 Both Statement I and Statement II incorrect.
- 2 Statement I is correct but Statement II incorrect.
- 4 Statement I is incorrect but Statement II incorrect.

Question: 49

Which of the following statements about inclusion bodies is incorrect?
[NEET-2020]

- 1 They are not bound by any membrane. T.
- 2 These are involved in ingestion of food particles. F.
- 3 They lie free in the cytoplasm. T.
- 4 These represent reserve material in cytoplasm. T.



Question: 50

Which of the following structures is not found in a prokaryotic cell?
[NEET-2015]

- 1 Mesosome
- 2 Plasma membrane
- 3 Nuclear envelope
- 4 Ribosome

Question: 51

Select the wrong statement.

[NEET-2016]

- 1 Bacterial cell wall is made up of peptidoglycan T.
- 2 Pili and fimbriae are mainly involved in motility of bacterial cells. X
- 3 Cyanobacteria lack flagellated cells T
- 4 Mycoplasma is a wall-less microorganism T.

Question: 52

The structure that help some bacteria to attach to rocks and/or host tissues are-
[NEET-2015]

1 Mesosomes

X

2 Holdfast

X

3 Rhizoids

X

4 Fimbriae

✓

2
 $45 \times 4 = 180$
Marks.

Thank
You